

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IInd - Semester of B.Pharm Examination

University Theory Examination October 2015

PH119 - Pharmaceutical Chemistry- II

Date: 14.10.2015, Wednesday Time: 10.00 a.m. to 1.00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION - I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION - I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION - I is 30 minutes. Answers to SECTION - I must be written during the first 30 minutes of the examination.
3. Answers to SECTION - II and SECTION - III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. How the organic compounds with molecular formula of $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3OCH_3 are related?
[A] isoelectronic species
[B] isotopes
[C] isomers
[D] These compounds are not related at all...they are totally different.
2. What is the total number of sigma bonds found in the following compound?
 $\text{CH}_3-\text{CH}=\text{C}=\text{CH}-\text{C}\equiv\text{C}-\text{H}$
[A] 8
[B] 10
[C] 11
[D] 15
3. What is the IUPAC name for the following compound?

$\begin{array}{c} \text{CH}_3 \qquad \qquad \text{CH}_3 \\ | \qquad \qquad \quad | \\ \text{CH}_3-\text{C}-\text{CH}_2-\text{CH} \\ | \qquad \qquad \quad | \\ \text{CH}_3 \qquad \qquad \text{CH}_3 \end{array}$

[A] 1,3-pentamethylpropane
[B] 1,1,3,3-tetramethylbutane
[C] 2,4,4-trimethylpentane
[D] 2,2,4-trimethylpentane
4. How many dichlorinated isomers can be formed by the halogenation of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ with Cl_2 in the presence of light?
[A] 2
[B] 3
[C] 5
[D] 6
5. Which of the following would not be a reasonable nucleophile in a $\text{S}_\text{N}2$ reaction?
[A] NH_3
[B] CN^-
[C] H_2O
[D] OH^-
6. The reason that CH_3NH_2 is a stronger base than CH_3OH is due to the fact that...
[A] hyperconjugation resonance of the product, CH_3NH_3^+ , is better than for CH_3OH_2^+ .
[B] resonance of the product, CH_3NH_3^+ , is better than for CH_3OH_2^+
[C] the electronegativity of nitrogen is less than oxygen, thus making the electron pair more available.
[D] It isn't. The oxygen atom has 2 non-bonded pairs of electrons making it twice as basic.
7. The hybridization of the central carbon in $\text{CH}_3\text{C}\equiv\text{N}$ and the bond angle CCN are

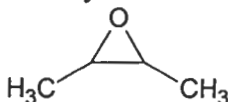
[A] sp^2 , 180°

[B] sp , 180°

[C] sp^2 , 120°

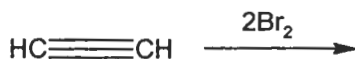
[D] sp^3 , 109°

8. The fifth member of the alkane series has the molecular formula _____.
- [A] C_5H_5
[B] C_5H_{12}
[C] C_5H_{10}
[D] C_5H_6
9. For which mechanisms the first step involved is the same?
- [A] $E1 + E2$
[B] $E2 + SN2$
[C] $E1$ and $SN1$
[D] $SN1$ and $SN2$
10. All statement is correct for SN_1 reaction. Except one
- [A] Follow first order kinetic
[B] Rearrangement is possible
[C] Inversion of configuration takes place
[D] Two step reaction
11. The ether used in Wurtz synthesis is...
- [A] Dry
[B] Acidic
[C] Basic
[D] Aqueous
12. Which of the following is a correct formula of Grignard reagent?
- [A] $RMgX$
[B] RMg_2X
[C] $RMgX_2$
[D] All
13. IUPAC name of *iso*-hexane is...
- [A] 2-methyl pentane
[B] 1,1-dimethyl butane
[C] 2-methyl hexane
[D] 1,1-dimethyl pentane
14. Identify the basic ring system in following structure:

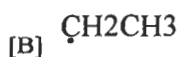
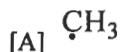


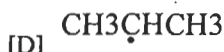
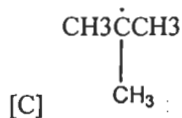
- [A] Oxirane
[B] Aziridine
[C] Pyridine
[D] Oxazole

15. What is the product of following reaction?



- [A] 1,2 DibromoEtahne
[B] 1,1',2,2' Tetrabromo Ethane
[C] 1,2,2' Tri Bromo Ethane
[D] 2,2' Di bromoEtahne
16. Orientation of Elimination reaction follows...
- [A] Markoniov's rule
[B] Saytzeff rule
[C] Micheal addition
[D] Hoffmann rule
17. Which of the following is a polar protic solvent?
- [A] Ether
[B] Ethanol
[C] Chloroform
[D] All
18. Which of the following is the most stable carbocation?

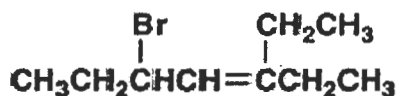




19. Unit of Dipole Moment is: _____

- [A] cm^{-1}
 [B] Poise
 [C] Debye
 [D] ppm

20. Which is the correct IUPAC name of the following compound?



- [A] 3-ethyl-5-bromohept-3-ene
 [B] 5-bromo-3-ethylhept-3-ene
 [C] 3-bromo-5-ethylhept-4-ene
 [D] 1,1-diethyl-3-bromopent-1-ene

SECTION – II

Q 2 Attempt any **FOUR** of the following

- | | | |
|---|--|----|
| A | Explain Inter molecular forces & Intra molecular forces with example. | 05 |
| B | Explain mechanism of Hoffmann degradation of amides. | 05 |
| | Define following terms: | |
| C | a) Nucleophile b) Dipole Moment c) Electrophile d) Free Radical | 05 |
| | e) Melting Point | |
| D | Write a short note on inductive effect. | 05 |
| E | Justify with suitable examples: E1 reactions are accompanied by rearrangement. | 05 |
| F | Add a short note on Sandmeyer reaction. | 05 |

SECTION – III

Q 3 Attempt any **FOUR** of the following

- | | | |
|---|---|----|
| A | State Markovnikov's rule, Anti-Markovnikov's rule and Saytzeff's rule and explain any two with suitable examples. | 05 |
| B | Give difference between SN1 and SN2 reactions. | 05 |
| C | Write short note on chemistry, preparation and properties of amines. | 05 |
| D | State and explain in detail about resonance. | 05 |
| E | Write two general methods of preparation and two reactions of alkanes and alkenes. | 05 |
| F | Give brief note on factor affecting elimination reaction. | 05 |

Q 4 Attempt any **FOUR** of the following

- | | | |
|---|--|----|
| A | Define and write short on preparations and properties of haloalkanes. | 05 |
| B | Define and explain hybridization and give detail note on sp^3 hybridization. | 05 |
| C | Draw structure for the following organic compounds: | 05 |
| | (i) 3-chloropentane (ii) 1-bromo-2-hydroxy cyclohexane | |
| | (iii) Benzylalcohol (iv) Methyl ethyl ether | |
| | (v) 3-bromo pent-1-ene | |
| D | Define and give two preparation and properties of carbanions. | 05 |
| E | Give brief note on quantitative analysis of nitrogen by Kjeldahl's method. | 05 |
| F | State and explain Saytzeff and Hoffmann elimination with example. | 05 |